

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Smart Lender – Loan Eligibility Prediction System** is a machine learning-powered web application designed to automate the loan eligibility assessment process. Traditional loan approval involves manual verification and lengthy decision-making, which may lead to delays and inconsistencies. This project leverages a trained **Random Forest classifier** integrated with a **Flask web application** to provide fast, accurate, and reliable loan eligibility predictions. The solution also incorporates secure user authentication, prediction history management, dashboards, and reporting features to improve operational efficiency and user experience.

Project Overview	
Objective	Develop a secure, intelligent, and user-friendly web application that predicts loan eligibility using machine learning while providing user authentication, role-based access control, prediction history, and administrative dashboards.
Scope	The project covers data preprocessing, exploratory data analysis, machine learning model training and evaluation, Flask web application development, SQLite database integration, secure user authentication, role-based authorization, prediction history management, dashboard implementation, and deployment for real-time loan eligibility prediction.
Problem Statement	
Description	Traditional loan approval systems rely heavily on manual verification and predefined rules, making the process time-consuming, inconsistent, and prone to human error. Financial institutions require an automated solution capable of evaluating applicant information quickly while maintaining accuracy, security, and transparency.
Impact	Implementing an automated loan eligibility prediction system improves operational efficiency, reduces manual effort, enhances prediction consistency, shortens processing time, and provides a better experience for both applicants and financial institutions.

Proposed Solution	
Approach	Develop a Flask-based web application integrated with a trained Random Forest machine learning model to analyze applicant information and predict loan eligibility. The application includes secure authentication, database management, prediction history tracking, dashboards, and reporting capabilities for efficient loan processing.
Key Features	☐ Machine Learning-based Loan Eligibility Prediction using Random Forest ☐ Secure User Registration and Login ☐ Role-Based Access Control ☐ Real-Time Loan Prediction ☐ Data Preprocessing and Feature Engineering ☐ Prediction History Management ☐ Interactive Dashboards ☐ CSV Report Export ☐ Responsive Web Interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Multi-core processor for model training and web application execution	Intel i5/Ryzen 5 or higher
Memory	RAM required for application and model execution	8 GB
Storage	Storage for datasets, trained models, and application files	1 TB SSD
Software		
Frameworks	Backend Web Framework	Flask, Flask-SQLAlchemy
Libraries	Machine Learning & Data Processing	Scikit-learn, Pandas, NumPy, Joblib, XGBoost, Matplotlib
Development Environment	IDE & Version Control	Visual Studio Code, Jupyter Notebook, Git, GitHub
Data		
Training Dataset	Loan Eligibility Dataset	614 records (CSV)
Testing Dataset	Loan Eligibility Dataset	368 records (CSV)
Database	User Accounts & Prediction History	SQLite Database